Diagnostic Translatability Between Ethnic Populations for Eczema

Introduction

Eczema, often referred to as atopic dermatitis (AD), is a chronic skin condition whose symptoms include itchiness, dry skin, rashes, scaly patches, blisters and skin infections[1]. Approximately 31.1 million Americans (10.1% of the U.S. population) have AD, and the disease can show up anywhere between childhood and adulthood. Symptoms can have a wide range of intensity, as its effects differ from person to person. Eczema symptoms can also flare-up or go into remission depending on environmental triggers.

There currently is no definitive cause for eczema, but factors often attributed to its development are genetics and the environment. For genetics, family history indicates a higher likelihood of developing eczema. Additionally, a specific gene known as filaggrin (FLG) is often associated with eczema, as mutations in the FLG gene cause skin barrier dysfunction, leading to excessive water loss and increased susceptibility to environmental triggers[2]. Besides these genetic components, there are certain environmental conditions related to eczema development. Several studies have found an association between new eczema cases and long-term exposure to air pollution, and another study found that exposure to isocyanate induced eczema in mice[3].

In order for an individual to be diagnosed with eczema, a healthcare provider, preferably a dermatologist, would need to examine your skin, and additional tests such as a skin biopsy or patch testing may be required[4,5]. However, not all symptoms appear similarly on all patients. For example, the typical "red, dry, and itchy rash" is easily seen on White populations, but Black populations do not express this symptom as saliently[6]. While there are many symptoms to consider, such differences can lead to underdiagnosis based on ethnic backgrounds. And while a professional dermatologist may be able to reduce this discrepancy in diagnoses, minority populations generally do not have as high access to healthcare as White populations as a result of systemic racism.

As of now, there is no cure for eczema. Eczema treatment is centered around symptom management, having goals such as reducing inflammation, preventing flares, etc[7]. The most common type of eczema treatment is through medication, and the recommended method depends on the severity of the symptoms. For example, moisturizing creams can be used to restore or protect the skin barrier while corticosteroid creams and calcineurin inhibitors work to decrease skin inflammation and prevent flares.

There has been some research on eczema prevention, but so far there has only been modest improvements[3]. Most experiments involve pregnant, breastfeeding, or infant individuals, and the interventions involve early treatment, supplementation, or exposure avoidance. The aim of this research is to improve the skin barrier at an early stage to reduce risk of eczema later in development.

When an individual has eczema, it has been shown that their QoL scores decrease by a significant amount[8]. Itch and sleep-related issues are strongly associated with eczema, leading to an overall more irritable mood. There are also psychological comorbidities with eczema. Mental disorders, psychological development, and strained social relationships are often seen in patients with eczema, and high levels of anxiety and depression have been found not only in the diagnosed individual, but also the parents due to the extra work that comes when dealing with eczema. Outside of these psychological consequences of eczema, common comorbidities include asthma, hay fever, food allergies, and cutaneous infections.

Because of diagnostic issues, QoL score drops, and existing symptom treatment for eczema patients, we attempted to develop a model to classify individuals with or without eczema using data from AllofUs controlled tier database with the goal of providing a CDS that can assist with the diagnostic process and get patients with eczema earlier treatment. However, the AllofUs controlled tier database had a significantly larger White population than minority populations. With the available data, the predictive model was solely trained on the White population, but tested on both White and Black populations. The reason for this is to determine if a model can be used to assess an underrepresented population's risk of developing eczema based on data from a White population and potentially highlight genetic causes of eczema based on ancestral history.

Literature Review

There are many disparities between Black and White populations for eczema, whether that be disease prevalence, diagnosis rates, access to care, etc. Croce's article[9], "Reframing racial and ethnic disparities in atopic dermatitis in Black and Latinx populations" covers these disparities, and it is for many of these reasons that we believe a predictive model would serve to be useful in getting people the treatment they deserve and close the disparity.

One of the main points in Croce's article is that environmental factors have a much larger effect on eczema prevalence than genetic factors. One such reason is because Black children are approximately 1.7-2.1 times more likely to receive an eczema diagnosis than White children. However, genetic polymorphisms associated with eczema are less common among Black people. These two statements imply that environmental factors hold more weight, and that would prove to be helpful in making a predictive model for both White and Black populations, as genetic differences would not hold strong influence in the model. Other analyses on FLG mutation prevalence, transepidermal water loss (TEWL), and immunologic phenotypes all support the idea that genetic factors do not explain the difference in eczema prevalence across races.

Croce then moves on to discuss contextual factors that influence eczema prevalence, severity, and persistence. First, they note that higher SES tends to be associated with increased eczema prevalence, but lower SES is associated with increased severity and persistence of eczema. The explanation for this pattern is that higher SES increases risk of developing eczema, but better access to care helps reduce the severity and persistence of the disease. They also hypothesize that higher SES may lead to overrepresentation in studies that connect SES and eczema. In addition to SES, Croce attributes eczema to a variety of environmental exposures, including air pollution, chemical exposures, allergens, and psychosocial stress, all of which has been associated with eczema in other literature.

Lastly, Croce touches on health care disparities for eczema. They state that Black children are significantly less likely to see a dermatologist for eczema, but are also 3 times more likely to receive an eczema diagnosis during that visit. They also mention that Black and Latinx children are 3 times more likely to have a health visit for eczema when compared to White children. These differences in health care highlight the need for a sufficient diagnostic tool, as there are many missed diagnoses in minority populations that can lead to greater healthcare resource consumption and QoL drops.

Another article that covers similar topics is Mosam's article[10] "Global epidemiology and disparities in atopic dermatitis". Most of the content reaches similar conclusions as Croce's paper, but Mosam delves a bit deeper into global disparities and underdeveloped eczema guidelines.

Mosam claims that an ideal set of diagnostic criteria is needed in order to garner meaningful epidemiology results for eczema and states that diagnostic criteria is not sufficient for non-white populations. For example, very few countries in South America and Africa have set guidelines for eczema diagnosis, and there is significant underrepresentation of these people in these countries. With a lack of standards, there is wide variation in eczema prevalence and severity, decreasing the significance of such findings. Not having set guidelines will also cause some individuals to be misdiagnosed, leading to insufficient treatment. Once again, development of a predictive model would prove beneficial for both White and minority populations as it would offer a tool to help with eczema diagnoses and bring medical attention to the people who need it.

Data and Methods

The dataset used in this analysis is obtained from the AllofUs controlled tier database. Since the goal of the analysis is to determine whether a predictive model for eczema prevalence in a White population can be translated to a Black population based on demographic and medical history data, the first cohort constructed was people who suffer from eczema, totaling up to 16,460 persons. Of those, 10,108 persons are White, 2,880 are Black or African American, and the rest are categorized into other races. The imbalance of White dataset availability is clearly seen here and will provide an interesting analysis. After the cohort was set up, the dataset was constructed using the cohort, where the concept sets selected for people suffering from eczema are: Demographics, Survey responses (The Basics, Lifestyle, Overall Health, Healthcare Access and Utilization, Social Determinants of Health, Personal and Family History), Fitbit Heart Rate Summary, Fitbit Heart Rate Level, Fitbit Activity Summary, Fitbit Intra Daily Steps, Fitbit Daily Sleep Summary, and Zip Code and Socioeconomic Status Data. Lastly for this dataset, all possible columns are selected to maximize the possible number of features to include. Since the model will aim to classify whether each person in the dataset suffers from eczema, a dataset of negative values (people without eczema) was retrieved. For this dataset, the same concept sets and columns are selected. However, this is extracted from all the participants as opposed to exclusively people suffering from eczema for the previous dataset. The resulting cohort is a group of 413359 person_ids.

The code to retrieve all these datasets was run in the Python environment, and several data processing steps were done. First, the survey data has multiple responses from a single person, separated by the type of survey they answered and the date of response. To compress the data into a processable form, the data was grouped by the person_id and survey column by the most frequent response, effectively “unmelting” the dataset into a horizontal configuration. Missing values are replaced by ‘unknown’ to flag the data entry as a distinct class instead of tagging it as missing with Nan. Meanwhile, Fitbit data was handled in a different manner. Since the Fitbit data contains multiple data points for a person_id, the dataset is grouped by person_id to create a new dataframe that has columns for different summary statistics of each original column. For instance, the “very_active_minutes” feature from the retrieved Fitbit activity dataframe is split into the mean, min, and max of the very active minutes, grouping the data by person_id but preserving the indicative statistics. Similarly, this type of processing was done on a majority of the Fitbit dataset columns, which are selected based on the predicted relationship or significance with the classifier. This type of data cleaning was replicated for the dataset containing all the participants, resulting in multiple dataframe for both eczema and all-participants datasets.

After the datasets were processed to cleaner and more manageable dataframes, the demographic, location, survey, and Fitbit data are merged into a single dataframe by the person_id. The remaining missing data was replaced by -1 if they were a quantitative variable, or with 'unknown' if they were strings. This was done to flag the missing data as a special case, indicating to the model that it belongs to a distinct category. Similarly, the all-participants dataset was also merged together in an identical fashion.

Finally, after both cohorts' dataframes were engineered into manageable formats, an additional column was added to the datasets. The new 'Eczema' column contains 0 for all rows in the all-participants dataframe, and 1 in the eczema cohort dataframe, which classifies whether the person_id suffers from eczema. The two dataframes were then joined together by first appending the two dataframes into each other, sorting by the 'Eczema' column, and finally dropping duplicate person_ids. As the command prioritizes the row of person_id further up in the dataframe, the "1" eczema classification is prioritized, effectively replacing all the eczema classifications with the correct one in the final dataframe.

The combined dataframe was then filtered into a dedicated dataframe for participants who answered "white" under race. After studying the dataframe, some columns required additional feature engineering. The participant's date of birth was first converted into a datetime format and redundant columns such as race_concept_id, gender_concept_id, and several others are dropped, as these are just numerical identifiers for the categorical data that it corresponds to. At this stage, the dataframe contains both numerical and categorical data. Therefore, One Hot Encoding was performed on the categorical data to convert it into numerical data processable by the model later on. This identical procedure was also applied to a dataframe that exclusively contained the Black or African American population.

To prepare the whites-dataframe for model training and testing, the dataset was separated into X and y, which have all the features and the 'Eczema' classification column from the whites-dataframe respectively. These two dataframes were then split into training and testing data using scikitlearn's train_test_split function, splitting the rows by 80% training data and 20% testing data. The model used in this case is scikitlearn's RandomForestClassifier, with class_weight = 'balanced', to help balance the initially imbalanced amount of positive and negative eczema classification. After fitting the model and testing on the test dataset, the model was validated with a high accuracy score. The same model was then applied to the black population dataframe, also producing a high accuracy score.

To make the analysis more practical, we attempted to reduce the required amount of features used to train the classifier. To achieve this, a feature importance dataframe was extracted from the model, ranking the features from the most important to the least according to the model, and selecting the top 20 features. The whites-dataframe and blacks-dataframe were then filtered to only contain the top 20 features. Similar to how the model was generated previously, the whites-dataframe was separated into train and test sets with an 80% to 20% ratio, and a model was fitted with the truncated dataframe. A similar high accuracy score was achieved, confirming that the model relies much more heavily on the top features than the rest. Afterward, the same model was used to predict the results from the truncated blacks-dataframe. Comparison between the actual and predicted values with a confusion matrix also confirms that the high accuracy score was preserved across the two populations.

Repository for code: https://github.com/vincent-angelo/CBB-740

Results

The results for all the model predictions are presented in a confusion matrix, where the predicted classification is compared with the actual classification to understand how the model performs and see the rate of false positives and negatives.

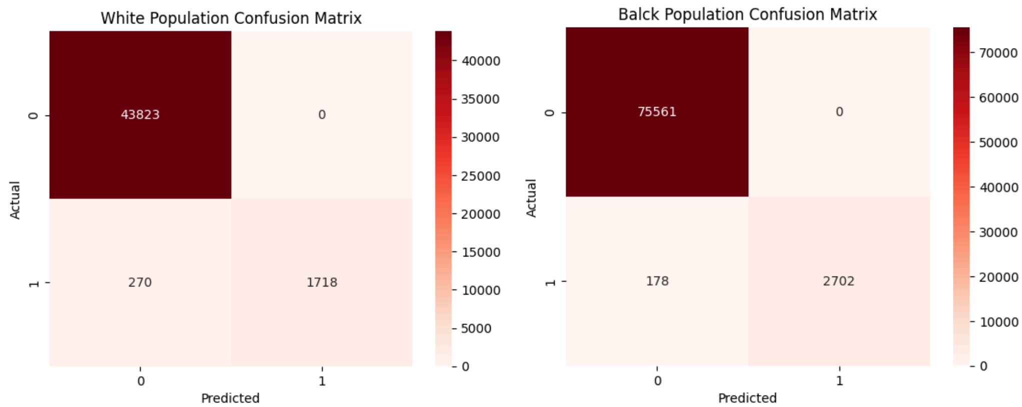

Figure 1A. Confusion matrix for the model trained and tested with the whites dataset (left), and confusion matrix for the same model tested on the blacks dataset (right).

The confusion matrix displayed above shows how the model trained on the white dataset is applicable to the black population dataset as well, achieving a high rate of true positives and negatives. The presented confusion matrix shows a disparity in dataset proportions: the matrix for the White population is made from a sample representing 20% of the White dataset, meanwhile the matrix for the Black population was made using the entire Black dataset, highlighting the discrepancy in the total data volumes between the groups.

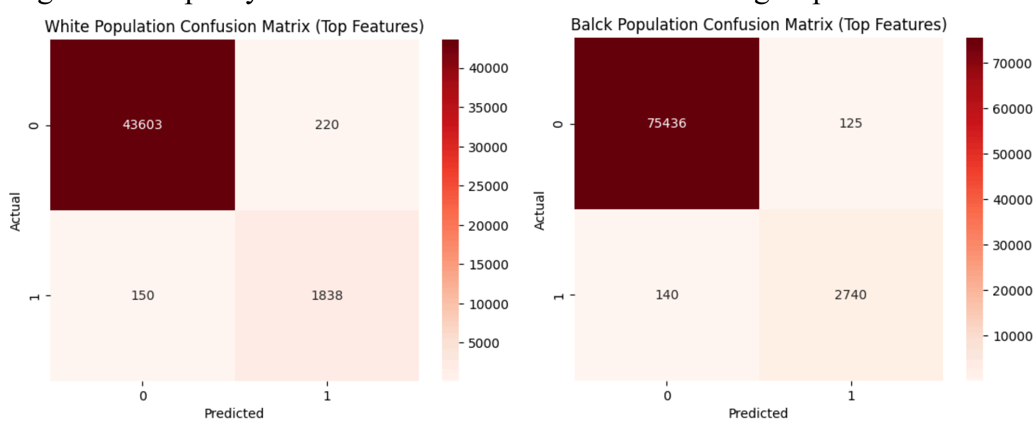

Figure 1B. The confusion matrix for the model trained and tested with the whites dataset was filtered to only include the top 20 features (left), and the confusion matrix for the same model tested on the blacks dataset with only the top 20 features included (right).

To further analyze the model, the importance of each feature to build the model was ranked, and the dataframe was filtered to only include the top 20 features. These features would then be used to test how robust the model will still be with a limited number of features. The resulting confusion matrix confirms that the model functions well with only the top 20 features, as listed in Figure 1C.

Feature	Importance
Personal and Family Health History_PMI: Skip	0.367
Personal and Family Health History_unknown	0.262
Personal and Family Health History_PMI: None	0.061
year_of_birth	0.050
Personal and Family Health History_Who in your family has had sudden death? - Grandparent	0.021
Personal and Family Health History_Who in your family has had sudden death? - Father	0.017
Personal and Family Health History_Who in your family has had sudden death? - Mother	0.008
Personal and Family Health History_Sibling Kidney Condition: Kidney Disease	0.008
Personal and Family Health History_Who in your family has had sudden death? - Sibling	0.007
Personal and Family Health History_PMI: Dont Know	0.007
zip_code_535**	0.006
vacant_housing	0.006
Overall Health_Social Satisfaction: Very Good	0.005
no_health_insurance	0.005
zip_code_537**	0.005
Lifestyle_Which Drugs Used: Marijuana Use	0.005
assisted_income	0.005
high_school_education	0.005
poverty	0.005
Overall Health_Social Satisfaction: Good	0.005

Figure 1C. Ranking and importance of the features used in the random forest classifier model trained on the white population dataset.

From Figure 1C, it is apparent that the absence of data from the personal and family health history plays an important role in the shaping of the model. These features originate from the Personal and Family Health History survey responses in the survey dataframe, and the Personal and Family Health History_unknown feature is the artificial data flag mentioned in the data and methods section. To understand the importance of these absences, another random forest model was tested without the top 3 of the features above, resulting in the confusion matrix shown in Figure 1D. Besides that, demographic data also claims a majority of the spot in the top 20 features, with a few zipcode data. From the table, it appears that the Fitbit data are not significant enough for the model to make accurate predictions.

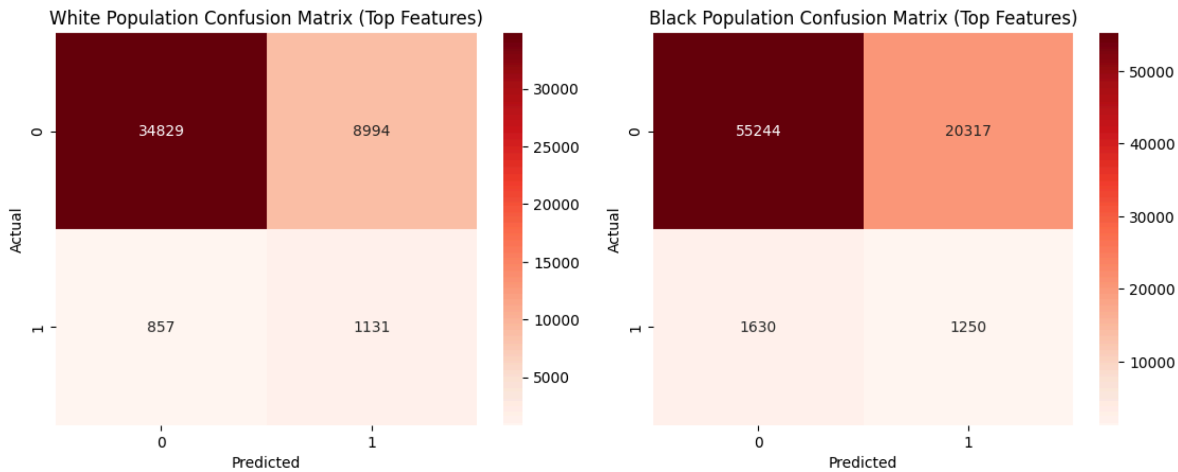

Figure 1D. Confusion matrix of the fourth to the 20th most important features.

As shown in Figure 1D, the model produced significantly less accurate predictions, with a large increase in both the false positive and negative predictions. Therefore, the missing data flags are an important consideration to the model's accuracy and should not be excluded.

Discussion

Looking at the predictive model generated using all the available features in the dataset, we were able to have very high diagnostic accuracy for both White populations and Black populations. What this implies is that genetic factors for eczema do not sufficiently explain the difference in eczema prevalence between ethnicities, and that environmental factors are much more likely to influence the development of eczema. This finding coincides with Croce's and Mosam's articles. With such an accurate diagnostic model for eczema, it shows that given enough contextual features we can provide a strong diagnostic tool that can help properly diagnose minority populations for eczema.

This model becomes more promising when it is reduced to only 20 features. When using 20 features to train and test the model, there is a similar level of accuracy in its diagnoses and the features themselves do not require any genetic testing or blood work. This means it can be used for diagnostic purposes, but also has the potential to generate risk scores to monitor patients for eczema development.

We had also decided to train and test a model without the top three features because they do not provide concrete answers to a patient's background. Unfortunately, this had lowered the accuracy of the model significantly, indicating that missing data on personal and family health history is important to eczema diagnosis. One possible interpretation of this result is that individuals without private medical insurance or incomplete family history are less inclined to fill out this data, and that may influence the quality of healthcare they receive. And as was shown in Croce's and Mosam's papers, eczema prevalence, severity, and persistence are influenced by an individual's SES. Outside of the top three features, most of the remaining top 20 features focus on SES factors. The questions regarding sudden death under personal and family health history can be indicative of lower access to healthcare. The importance of health social satisfaction has also been shown to increase eczema prevalence, but not necessarily severity and persistence (as was mentioned in the Croce article).

One particular feature that should be pointed out is the zip code. In the method used to train and test the predictive models, one-hot encoding was used on the zip codes to make it easier to work with. However, this led to zip codes as a whole appearing to have lower feature importance since each zip code was its own binary classification. In spite of this, within the top 20 features zip codes 535** and 537** are listed. Both 535** and 537** are located around Madison, Wisconsin. This indicates that geographic location is important to eczema diagnosis, and its importance should actually be higher due to the use of one-hot encoding.

Overall, the results of the various models that were built show that an assistive diagnostic tool can be made to help with accurate eczema diagnosis without relying on genetic factors. It also opens up the idea of making a model to calculate someone's risk of developing eczema which may prove helpful if more research is done on preventative methods in eczema.

Ethical, Legal, and Policy Considerations

Data privacy is one concern that should be addressed. Even when considering only the top 20 features obtained from our model, it is important that personal and family health history remain secure. Zip code data must also be kept safe, but this may hinder the development of a reliable model that does not rely on missing responses for eczema diagnosis. Having the full zip code would provide valuable data in determining where eczema is more prevalent and what associated features in the environment may influence its development, but that would also pose a security risk for individual's whose data was gathered. While zip code data is the first example of protected data that comes to mind for this dataset, it is important to take similar precautions for the other features, stripping identifying patient data and following the guidelines set by the AllofUs controlled tier database.

In addition to data privacy, we must make sure that a model would be resistant to bias. While we were able to obtain a model that worked for both Black and White populations when only trained on White populations, it would be beneficial to obtain more data on Black populations. By generating a model trained on Black populations, it can further validate the general use of the model trained on White populations. Keeping an eye out for biases between populations is key since the goal is to shrink the disparities found in eczema patients.

Looking towards real world application of these findings, eczema diagnosis can be more accurate and easier for both the patient and healthcare provider. The data used in our model can be obtained through in person health visits or over online questionnaires. This can prompt follow ups to confirm the presence of eczema, allowing more people to receive the appropriate treatment.

Contributions

Vincent Angelo (vma7) - processed data, wrote data and methods, results, and created figures
Andrew Yu (ajy28) - wrote introduction, literature review, discussion, and ethical, legal, and policy considerations

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